

Min-En (Brian) Wu

Los Angeles, CA • (424) 483-9071 • minenbrian@gmail.com • github.com/bw493

Education

University of California, Los Angeles | B.S. Electrical Engineering

Sep 2024 – Expected Jun 2028

National Taiwan University | College Coursework

Feb 2023 – Jun 2023

Introduction to Biomedical Engineering, Dept. of Electrical Engineering (A+); Political Sociology (A)

Research Experience

Research Intern | Interconnected & Integrated Bioelectronics Lab, UCLA

Dec 2024 – Nov 2025

Advanced an ESP32-based computer vision pipeline for real-time magnet and droplet detection, integrating YOLO-SAM2 segmentation, multithreading, and latency analysis to achieve 60+ FPS on edge devices.

Added SAM2 binary masking for pixel-based droplet volume quantification on a 1,000-image training dataset, and extended tracking with TubeletGraph by configuring CropFormer and Mask2Former under CUDA 12.6 and resolving Detectron2 compatibility.

Redesigned a ferrobotic cartridge compatible with 28 liquid-phase assays, built a Selenium-BS4 test library, and wrote deterministic Python scripts converting off-chip ELISA protocols to on-chip.

Summer Research Intern | CITI, Academia Sinica

Jun 2024 – Aug 2024

Developed a VILT + BERT multimodal retrieval system returning products conditioned on user queries, and fine-tuned BERT and custom BERT variants for the TREC 2024 End-to-End Product Retrieval Competition.

Researcher | Dept. of Electrical Engineering, National Taiwan University

Sep 2021 – Apr 2024

Designed a dual-channel optoelectronic human cerebral NIRS phantom with a 3D-printed precision position holder, reducing measurement error by 83%. [Research Paper Draft](#)

Technical Projects

Independent Developer | Thermloop, Predictive Thermal Management Platform

Jul 2026 – Present

Built a C++ platform that forecasts temperature and cools proactively using the electro-thermal analogy, portable across ESP32-S3 (ADS1299 EEG), Raspberry Pi (PiEEG), and Arduino UNO R3 via OpenBCI GUI.

Implemented predictive thermal throttling through causal convolution against a precomputed impulse-response kernel, so each cooling decision acts before overheating rather than after. [GitHub Repository](#)

Independent Developer | LTspice Circuit Simulation Automation via MCP

Jun 2026 – Jul 2026

Engineered a Python/Bash daemon that runs native macOS LTspice binaries from structured AI instructions, bridged to Claude Desktop with an mcp-remote network bridge through macOS sandboxing.

Validated autonomous AC sweeps (10 Hz to 1 MHz) and transient analysis (0 to 1 ms) on an RC low-pass filter, tracking phase response and capacitor charging ($\tau \approx 0.1$ ms).

Leadership & Technical Experience

Scientist | CruX UCLA

Dec 2025 – Present

Built an MNE-Python pipeline with FFT-based spectral analysis to isolate beta-band power, resolving epoch segmentation failures from missing PsychoPy-to-OpenBCI event markers.

Designed a beta-band (13–30 Hz) EEG neurofeedback protocol with a 100-trial go/no-go task across four participant sessions, recording on an OpenBCI Cyton at frontal (F3, Fz, F4) and central (C3, Cz, C4) sites.

Analyzed task accuracy and reaction time to set a control baseline for neurofeedback comparison.

External Outreach Officer | CruX UCLA

Mar 2026 – Present

Arrange speakers, networking events, and workshops; manage outreach and member internship opportunities.

Work Experience

Intern | AppWorks, Startup Accelerator and Venture Capital

Aug 2023 – Jan 2024

Evaluated AI and Web3 startups in Southeast Asia for venture investment and presented reports to analysts.